HOSSAM EMAD MOHAMED OSMAN

Embedded Systems / Hardware Engineer

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PROFESSIONAL SUMMARY

Embedded Systems Engineer with 2+ years of hands-on experience in hardware design, PCB layout, firmware development, and production support. Proven ability to independently identify field issues, propose solutions, and deliver improved hardware and firmware revisions end-to-end. Experienced in automotive and industrial IoT systems, 4-layer high-speed PCB design, EMI mitigation, controlled impedance routing, BMS, and RTOS-based firmware — with full product lifecycle ownership from concept through manufacturing and deployment.

TECHNICAL SKILLS

Firmware & Software: Embedded C (production-grade), Python (prototyping & validation), RTOS (FreeRTOS), SMS/MQTT/HTTP, GPS data processing, FOTA, Geofencing algorithms, Embedded debugging, Root-cause analysis

Hardware & PCB: Multi-layer PCB design (up to 4 layers), Stackup planning & controlled impedance routing (USB, CAN, RF), High-speed design, EMI mitigation (ground stitching, return path optimization), RF subsystem design (antenna placement & impedance matching), Power electronics, Battery Management Systems (BMS — 3-cell), Board bring-up, soldering & field debugging, Full manufacturing file preparation (Gerber, BOM, CPL), Fab coordination (JLCPCB)

Protocols & Interfaces: CAN (High-Speed, SWCAN, scanning), USB, UART/SPI/I2C, RS232/RS485, Wi-Fi/BLE, 4G (SIMCom), ADC/GPIO

System & Product Engineering: Full product lifecycle ownership (concept → design → manufacturing → deployment), Reverse engineering & hardware redesign, Mechanical integration & enclosure selection, Automotive telematics & Industrial IoT gateways

Tools: Altium Designer (advanced schematic & layout, actively upskilling), EAGLE, EasyEDA Pro, KiCad, Autodesk Fusion 360, LTspice, JLCPCB manufacturing

PROFESSIONAL EXPERIENCE

Embedded Systems Engineer *Jul 2024 – Present*
SEITech Solutions (Acquired by T&S Group) — Smart Village, Cairo

SEIFence — IoT Tracking & Control Device (Hardware & Firmware Version 2)

- Identified hardware, mechanical, and firmware issues in Version 1 during early field deployment across ~200 units.
- Independently proposed, designed, and developed a complete improved Hardware Version 2 (4-layer PCB), reducing costs and enhancing reliability.
- Designed and implemented full RTOS-based production firmware in C: SMS/MQTT/HTTP, GPS processing, FOTA, and geofencing algorithm.
- Built the firmware stack twice — an initial Python-based version for functional validation, followed by an optimized, production-grade version in C.
- Resolved GPS antenna stability issues with post-fab PCB fixes (drilled holes + through-hole pads); successfully transitioned 120+ units to production Version 2.
- Provided ongoing aftersales hardware maintenance and debugging for all fielded units.

ForWheelz — Automotive Telematics Device

- Led full hardware development lifecycle for STM32-based automotive telematics ECU with 3 CAN interfaces, USB, RF, and power management.
- Designed 2-board stacked 4-layer impedance-controlled PCB with integrated RF subsystem (GSM antenna, matching network, RF routing constraints).

- Applied EMI mitigation techniques including ground stitching and return path optimization for automotive signal integrity.
- Managed compact enclosure integration under automotive constraints and coordinated full manufacturing package and supplier communication.
- Supported system validation and field optimization through deployment.

SEITech Box — Industrial IoT Gateway & Power Backup System

- Designed a 3-cell BMS and backup power module for the Edge Box ESP100 industrial gateway, providing seamless backup power during inverter outages.
- Implemented a controlled DC-DC power path with real-time charging/discharging monitoring; completed manual assembly, board bring-up, and full validation ahead of customer delivery.
- Reverse-engineered existing Edge Box hardware and redesigned it around an ESP32-S3 platform, integrating Wi-Fi, BLE, SIMCom 4G, GPS, RS232/RS485, ADC, GPIO, and power management circuits.
- Selected enclosure and completed schematic capture, PCB layout, and fabrication file preparation using KiCad.

Access Control Board — Industrial Ethernet & RS485 Remote I/O Controller

- Designed a production-ready embedded controller around an STM32F401RCT6 (84 MHz Cortex-M4) for industrial automation and remote monitoring applications.
- Implemented Ethernet connectivity (ENC28J60, RJ45) and an isolated RS485 interface (MAX3485 with optocoupler isolation) for reliable industrial networking.
- Integrated dual 24LC512 EEPROM devices over I^2C for non-volatile data logging, plus a USB Type-C programming/debugging interface with independent power.
- Developed a multi-rail power architecture (12V input, 5V via LM2596S, 3.3V via AMS1117 LDO) with comprehensive protection: TVS, ESD, over-voltage/surge suppression, and decoupling networks.
- Designed digital I/O monitoring and isolated relay-driving circuitry with multi-LED diagnostics; completed schematic capture, layout review, and BOM preparation.
- Delivered a complete production package: Gerbers, pick-and-place files, assembly drawings, LCSC-sourced BOM, and revision-controlled design archives.

Smart Wi-Fi Switch — ESP8266-Based IoT Smart Home Device

- Designed a 3-channel Smart Wi-Fi Switch enabling remote control of AC loads via an ESP8266 (ESP-12E), with isolated high-voltage AC and low-voltage control sections.
- Defined system architecture, selected components, and built/validated the initial prototype on an FR4 board.
- Completed schematic and PCB layout in KiCad, generated Gerber/BOM manufacturing files, and programmed firmware via UART.
- Performed hardware debugging, functional validation, and end-to-end system testing.

Tools: *EAGLE, EasyEDA Pro, KiCad, Fusion 360, LTspice*

Industrial IoT Controller & Hardware Development Platform

- Designed and developed a custom Industrial IoT controller based on STM32F407 and ESP32-S3, integrating Ethernet (ENC28J60), CAN, RS-485, USB Type-C, QSPI Flash, environmental sensors, and expansion interfaces.
- Designed the complete multi-sheet schematic, PCB layout, power architecture, and protection circuitry using KiCad.
- Implemented high-speed interfaces including Ethernet, CAN, RS-485, USB, SPI, and QSPI.
- Performed PCB fabrication review, assembly, firmware flashing, hardware bring-up, debugging, and system validation.

Embedded Systems Engineer *Jan 2024 – Feb 2024*

Noziqs Medical — Nasr City (Short-term contract engagement)

- Developed embedded firmware for operating tables, improving system functionality by 30%.
- Reduced device downtime by 20% via board-level troubleshooting and root-cause analysis.

EDUCATION

Bachelor of Electronics and Control Engineering *Sep 2017 – Jul 2022*

Menoufia University | Grade: Very Good

Note: Following graduation, pursued the NTI Automotive Embedded Systems Diploma (Feb–Jul 2024) before transitioning to industry.

CERTIFICATIONS & TRAINING

Crash Course Electronics & PCB Design — Udemy

Jan 2026

112 hours covering AC/DC analysis, analog/digital electronics, PCB design flow: schematic → layout → fabrication → bring-up.

Automotive Embedded Systems Diploma — NTI, New Capital

Feb 2024 – Jul 2024

Embedded C, RTOS, STM32 & AVR, CAN/LIN/UART/SPI/I2C, bootloader/FOTA, automotive testing, project management.

Hardware Design Mentorship

Professional PCB design (Altium Designer), ESP32-S3 & RP2040 dev boards, advanced grounding and layout practices.

LANGUAGES

English — Professional Proficiency